

1) Introduction: Machine learning (ML) and Deep Learning are two important branches of artificial intelligence (AI). Both enable computers to learn from data and make predictions or decisions without being explicitly programmed. Deep learning is a specialized subset of machine learning that uses artificial neural networks with multiple layers to solve complex problems.

Difference Between Machine Learning and Deep Learning

<u>Aspect</u>	<u>Machine Learning (ML)</u>	<u>Deep Learning (DL)</u>
<u>Architecture</u>	Uses simple algorithms such as Decision Trees, SVM, Naive Bayes and Random Forest	Uses multi-layered artificial neural networks (ANN) especially Deep learning
<u>Data Requirements</u>	Works well with small to medium sized datasets	Requires very large datasets for high accuracy
<u>Feature Extraction</u>	Features are manually selected and engineered by humans	Features are automatically learned from raw data by the neural network
<u>Applications</u>	Suitable for structured data like tabular datasets	Best for unstructured data such as images, audio and text

Real-world applications

1) Email spam detection - Classifies email as spam (or) non-spam

2. Credit Card fraud detection: Detects fraudulent financial transactions

Deep Learning:

1. Face Recognition System - Used in Smartphone face unlock and security system.

2. Autonomous Vehicles: Detects roads, pedestrians, traffic signals, other vehicles for self-driving cars.

Conclusion:-

Machine learning and Deep learning are both powerful AI technologies, but they differ in their architecture, data requirements, feature extraction methods, applications.

2) Introduction:-

Representation Learning is a technique in Deep learning where a neural network automatically learns useful features from raw data instead of relying on manually designed features. It enables the model to identify important patterns and representations that improve prediction accuracy.

Representation Learning:-

It is a deep learning technique in which a neural network automatically learns useful features.

Advantages

- Learns features automatically

- Reduces manual effort
- Improve performance on complex tasks like image and speech recognition

* Width of a Neural network:

Width refers to a number of neurons in each hidden layer

Influence:

- Learning Capability: Learns more features at the same level
- Computational Complexity: Requires more memory and computation

Depth of a Neural network:

Depth refers to the number of hidden layer in network

- Learning Capability: Learns simple to complex feature hierarchically

Comparison:-

<u>Aspect</u>	<u>Width</u>	<u>Depth</u>
Definition	Number of neurons in a hidden layer	Number of hidden
Learning	Learns more features	Learns complex hierarchical features
Complexity	Higher memory usage	Longer training time
Performance	Better Capacity	Better accuracy for tasks.

Conclusion :-

Representation learning enables neural networks to automatically learn features from data. width increases the number of neurons to learn more features, while depth increases the number of layers.

3) Introduction:-

Activation functions and non-linearity to neural networks, enabling them to learn complex patterns. ReLU, Leaky ReLU (LReLU) and ELU are commonly used activation function that differs in how they handles negative input values.

Comparison table:-

Feature	ReLU	LReLU	ELU
Behaviour	outputs 0 for negative values	Gives a small negative output	Give smooth negative output.
Advantages	Fast and Simple	Prevents dying ReLU	Faster Convergence, Better accuracy
Disadvantages	Dying ReLU Problem	Needs α value	More computation expensive
Use Cases	CNNs, Image classification	Deep neural networks	Image, Speech & NLP application

Mathematical Behaviour:

- ReLU : $f(x) = \max(0, x)$
- Leaky ReLU : $f(x) = x$ if $x > 0$, otherwise αx
- ELU : $f(x) = x$ if $x > 0$, otherwise $\alpha(e^{x-1})$

Conclusion:

- ReLU is simple and fast
- Leaky ReLU avoids the dying ReLU problem
- ELU provides smoother learning and better performance but requires more computation

4)

Introduction:-

unsupervised learning is a type of machine learning in which a neural network learns from unlabeled data. It finds hidden layers and relationships without knowing the output.

Restricted Boltzmann machine (RBM):

An RBM is a neural network with one visible layer and one hidden layer that learns the probability distribution of input data.

Working Principle:-

- Input is given to the visible layer
- Hidden layer extracts important features
- The network reconstructs the input and adjusts weights to reduce reconstruction error.

Applications:-

- Recommendation system
- Feature extraction

Autoencoders:-

An autoencoder is a neural network that learns to compress and reconstruct input data

Working Principle:-

- The encoder compresses the input
- The Decoder reconstructs the original input

Comparison:-

<u>Aspect</u>	<u>RBM</u>	<u>Autoencoder</u>
Objective	Learn data patterns	Reconstruct input data
Architecture	Visible + Hidden layer	Encoder + Decoder
Applications	Recommendation, feature extraction	Denoising, data compression.

Conclusion:-

Unsupervised learning helps neural networks discover hidden patterns from unlabeled data. RBM's are mainly used for feature learning and recommendation system.